

MA 102 (Mathematics II - Ordinary Differential Equations)

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Solutions of Tutorial Sheet No. 5

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Lectures 15-16: Systems of first-order equations, Matrix form of first-order systems

1. Represent each of the following equations in terms of an equivalent system of first-order equations:

(a) $x'' - t^2x' - tx = 0$, (b) $x''' = x'' - t^2(x')^2$, (c) $x'' - 3x' - 11x = \sin t$.

Further, write the derived systems in matrix form $\bar{x}' = A\bar{x} + \bar{b}$ (if possible).

Solution: (a) It can be written as a system as follows:

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= tx + t^2y.\end{aligned}$$

This system has the matrix form as follows:

$$\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ t & t^2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

(b) Take $\frac{dx}{dt} = y$ to get $\frac{d^2y}{dt^2} = \frac{dy}{dt} - t^2y^2$. Next, take $\frac{dy}{dt} = z$ to get the system as

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= z, \\ \frac{dz}{dt} &= z - t^2y^2.\end{aligned}$$

It is not possible to represent this system in $\bar{x}' = A\bar{x} + \bar{b}$ form since y appears nonlinearly.

(c) It can be written as a system as follows (non-homogeneous):

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= 11x + 3y + \sin t.\end{aligned}$$

This system has the matrix form as follows:

$$\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ 11 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ \sin t \end{bmatrix}.$$

2. If a particle of mass moves in the xy -plane, its equations of motion are

$$m \frac{d^2 x}{dt^2} = f(t, x, y), \quad m \frac{d^2 y}{dt^2} = g(t, x, y),$$

where f and g represent the x and y components, respectively, of the force acting on the particle. Express this system of two second-order equations by an equivalent system of four first-order equations.

Solution: It is easy to follow that the required system is

$$\begin{aligned} \frac{dx}{dt} &= u, \\ \frac{du}{dt} &= \frac{f(t, x, y)}{m}, \\ \frac{dy}{dt} &= v, \\ \frac{dv}{dt} &= \frac{g(t, x, y)}{m}. \end{aligned}$$

3. The vector functions $\bar{x}_1 = [e^{-t}, 2e^{-t}, e^{-t}]^T$, $\bar{x}_2 = [e^t, 0, e^t]^T$, $\bar{x}_3 = [e^{3t}, -e^{3t}, 2e^{3t}]^T$ are solutions to the system $\bar{x}'(t) = A\bar{x}(t)$. Determine whether they form a fundamental solution set. If they do, find a fundamental matrix for the system and give a general solution.

Solution:

Given $\bar{x}_1 = [e^{-t}, 2e^{-t}, e^{-t}]^T$, $\bar{x}_2 = [e^t, 0, e^t]^T$, $\bar{x}_3 = [e^{3t}, -e^{3t}, 2e^{3t}]^T$.

Now,

$$W(\bar{x}_1, \bar{x}_2, \bar{x}_3) = \begin{vmatrix} e^{-t} & e^t & e^{3t} \\ 2e^{-t} & 0 & -e^{3t} \\ e^{-t} & e^t & 2e^{3t} \end{vmatrix} = -2e^{3t} \neq 0.$$

Therefore, the given set of vectors is linearly independent and so it forms a fundamental solution set. A fundamental matrix is given by

$$\Phi(t) = \begin{bmatrix} e^{-t} & e^t & e^{3t} \\ 2e^{-t} & 0 & -e^{3t} \\ e^{-t} & e^t & 2e^{3t} \end{bmatrix}.$$

The general solution is given by

$$\bar{x}(t) = \Phi(t)\bar{c} = c_1 \begin{bmatrix} e^{-t} \\ 2e^{-t} \\ e^{-t} \end{bmatrix} + c_2 \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} + c_3 \begin{bmatrix} e^{3t} \\ -e^{3t} \\ 2e^{3t} \end{bmatrix}.$$

4. Find the general solution of the following first-order equations:

$$\begin{cases} \frac{dx}{dt} = 4x - y, \\ \frac{dy}{dt} = 2x + y. \end{cases}$$

Solution:

Given

$$\begin{cases} \frac{dx}{dt} = 4x - y, \\ \frac{dy}{dt} = 2x + y. \end{cases}$$

The auxiliary equation is $m^2 - 5m + 6 = 0$ giving the roots as $m_1 = 2, m_2 = 3$. The two linearly independent solutions are

$$\left. \begin{array}{l} x = A_1 e^{2t} \\ x = B_1 e^{2t} \end{array} \right\}, \quad \left. \begin{array}{l} x = A_2 e^{3t} \\ x = B_2 e^{3t} \end{array} \right\}.$$

Using the relation $(a_1 - m)A + b_1 B = 0$, we get $A_1 = 1, B_1 = 2, A_2 = 1, B_2 = 1$. Hence, the general solution can be written as

$$\begin{cases} x(t) = c_1 e^{2t} + c_2 e^{3t}, \\ y(t) = 2c_1 e^{2t} + c_2 e^{3t}. \end{cases}$$

5. Find the general solution of the following first-order equations:

$$\begin{cases} \frac{dx}{dt} = x + 3y, \\ \frac{dy}{dt} = 3x + y. \end{cases}$$

Then, find the particular solution given that $x(0) = 5$ and $y(0) = 1$.

Solution:

Given

$$\begin{cases} \frac{dx}{dt} = x + 3y, \\ \frac{dy}{dt} = 3x + y. \end{cases}$$

The auxiliary equation is $m^2 - 2m - 8 = 0$ giving the roots as $m_1 = -2, m_2 = 4$. The two linearly independent solutions are

$$\left. \begin{array}{l} x = A_1 e^{-2t} \\ x = B_1 e^{-2t} \end{array} \right\}, \quad \left. \begin{array}{l} x = A_2 e^{4t} \\ x = B_2 e^{4t} \end{array} \right\}.$$

Using the relation $(a_1 - m)A + b_1 B = 0$, we get $A_1 = 1, B_1 = -1, A_2 = 1, B_2 = 1$. Hence, the general solution can be written as

$$\begin{cases} x(t) = c_1 e^{-2t} + c_2 e^{4t}, \\ y(t) = -c_1 e^{-2t} + c_2 e^{4t}. \end{cases}$$

using the initial conditions $x(0) = 5, y(0) = 1$, we get $c_1 = 2, c_2 = 3$.

Therefore, the specific solution is given by

$$\begin{cases} x(t) = 2e^{-2t} + 3e^{4t}, \\ y(t) = -2e^{-2t} + 3e^{4t}. \end{cases}$$

6. Solve the given equation $\bar{x}'(t) = A\bar{x}(t)$ for

$$(a) A = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}, (b) A = \begin{bmatrix} -1 & 1 & -2 \\ 0 & -1 & 4 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution: The solution is given by

$$\bar{x} = \alpha e^{\lambda t} \bar{c}.$$

We have to find the values of λ from the characteristic equation $\det(\lambda I - A) = 0$ and the column vectors $\bar{c}$ from $(\lambda I - A)\bar{c} = \bar{0}$.

(a) The characteristic equation can be found as

$$(2 + \lambda)(\lambda^2 - 2\lambda + 2) = 0$$

We get the characteristic values as $\lambda = -2, 1 + i, 1 - i$.

Let $\bar{u}$ be the column vector corresponding to $\lambda = -2$. This can be obtained as

$$\bar{u} = [0 \quad 0 \quad 1]^T.$$

Let $\bar{v}$ be the column vector corresponding to $\lambda = 1 + i$. This can be obtained as

$$\bar{v} = [1 - i \quad -1 \quad 0]^T.$$

Let $\bar{w}$ be the column vector corresponding to $\lambda = 1 - i$. This can be obtained as

$$\bar{w} = [1 + i \quad -1 \quad 0]^T.$$

Hence, the general solution can be written as

$$\bar{x} = \alpha_1 \bar{u} e^{-2t} + \alpha_2 \bar{v} e^{(1+i)t} + \alpha_3 \bar{w} e^{(1-i)t}.$$

(b) The characteristic equation gives the characteristic values as $\lambda = -1$ (repeated) and $\lambda = 1$.

Let $\bar{u}$ be the column vector corresponding to $\lambda = -1$. This can be obtained as

$$\bar{u} = [1 \quad 0 \quad 0]^T.$$

Let $\bar{v}$ be another column vector corresponding to the same root $\lambda = -1$. This can be obtained as

$$\bar{v} = [t \quad 0 \quad 0]^T.$$

Let $\bar{w}$ be the column vector corresponding to $\lambda = 1$. This can be obtained as

$$\bar{w} = [0 \quad 2 \quad 1]^T.$$

Hence, the general solution can be written as

$$\bar{x} = \alpha_1 \bar{u} e^{-t} + \alpha_2 \bar{v} e^{-t} + \alpha_3 \bar{w} e^t.$$

7. Show that the solutions for the system

$$\begin{cases} \frac{dx}{dt} = x + 2y, \\ \frac{dy}{dt} = 3x + 2y. \end{cases}$$

are given by $\begin{cases} x = 2e^{4t}, \\ y = 3e^{4t}, \end{cases}$ and $\begin{cases} x = e^{-t}, \\ y = -e^{-t}. \end{cases}$

Further, show that the solutions of the above system can also be obtained by

(i) differentiating the first equation with respect to t and then eliminating y ,

(ii) differentiating the second equation with respect to t and then eliminating x .

Solution: For the first part, proceed exactly in the same manner as in Problems 4 and 5. Then,

(i) Differentiate the first equation of the system to get

$$\frac{d^2x}{dt^2} = \frac{dx}{dt} + 2\frac{dy}{dt}. \quad (1)$$

Further, subtracting the first equation of the system from the second:

$$\frac{dy}{dt} = \frac{dx}{dt} + 2x. \quad (2)$$

Eliminating $\frac{dy}{dt}$ between equations (1) and (2), we get the following second-order equation in x :

$$\frac{d^2x}{dt^2} - 3\frac{dx}{dt} - 4x = 0.$$

Its general solution is

$$x(t) = c_1e^{4t} + c_2e^{-t}.$$

(ii) Differentiate the second equation of the system to get

$$\frac{d^2y}{dt^2} = 3\frac{dx}{dt} + 2\frac{dy}{dt}. \quad (3)$$

Further, subtracting the second equation of the system from the first one multiplied by 3:

$$3\frac{dx}{dt} = \frac{dy}{dt} + 4y. \quad (4)$$

Eliminating $\frac{dx}{dt}$ between equations (3) and (4), we get the following second-order equation in y :

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} - 4y = 0.$$

Its general solution is

$$y(t) = c_1e^{4t} + c_2e^{-t}.$$

8. We know that, for a system of two first-order equations, the complex roots $a \pm ib$ of the auxiliary equations lead to the following solutions:

$$\left. \begin{aligned} x &= e^{at}(A_1 \cos bt - A_2 \sin bt), \\ y &= e^{at}(B_1 \cos bt - B_2 \sin bt), \end{aligned} \right\}$$

and

$$\left. \begin{aligned} x &= e^{at}(A_1 \sin bt + A_2 \cos bt), \\ y &= e^{at}(B_1 \sin bt + B_2 \cos bt). \end{aligned} \right\}$$

Find the Wronskian of these solutions. Check whether it is mandatory that $A_1B_2 - A_2B_1 \neq 0$.

Solution:

We have

$$\begin{aligned} W(x(t), y(t)) &= \begin{vmatrix} x_1(t) & x_2(t) \\ y_1(t) & y_2(t) \end{vmatrix} \\ &= \begin{vmatrix} e^{at}(A_1 \cos bt - A_2 \sin bt) & e^{at}(A_1 \sin bt + A_2 \cos bt) \\ e^{at}(B_1 \cos bt - B_2 \sin bt) & e^{at}(B_1 \sin bt + B_2 \cos bt) \end{vmatrix} \\ &= e^{2at}(A_1B_2 - A_2B_1). \end{aligned}$$

In order that the solutions are linearly independent, we must have $A_1B_2 - A_2B_1 \neq 0$.

The End